

PARAMETERS	N=81			
	value(%)	mean	median	range
SITE				
nose	26(32.1)			
cheek	19(23.5)			
ear	17(21)			
forehead	7(8.6)			
others	12(14.8)			
HISTOLOGY				
basal	55(67.9)			
squamous	23(28.4)			
mixed	2(2.5)			
bowens	1(1.2)			
STAGE				
I	68(84)			
II	13(16)			
VOL(CC)		1.8	1.3	0.3-9.4
# CATHETERS		5	5	1-13
V90 (%)		70.1	83.7	38.6-100
V135(%)		2.3	0.2	0.01-53

Mean follow up was 19 weeks (median 13 weeks, range 0-64), and 78 of the 81 lesions received the full treatment course. The complete response rate at last follow-up was 95%(77/81). Local failure was seen in 3.7%(3/81) and regional failure was seen in 1.2% of 81 lesions. Three patients with 7 lesions were lost to follow-up. The KM local relapse-free survival was 88.4% and loco-regional relapse-free survival was 83.5% at 64 weeks. There were no Grade 3 or Grade 4 toxicity, 49% patients had Grade 2 dermatitis and 51% had Grade 1 skin changes. There was complete resolution of skin toxicity by 3 months. One patient developed a persistent ulcer after subsequent biopsy for suspected recurrence.

Conclusions: 3-TAB is feasible treatment for HN skin cancers, with low acute toxicity. Early outcomes show good local control and low long term toxicity, but longer follow up is warranted.

PO-409

LOCAL EXCISION AND BRACHYTHERAPY FOR SPHINCTER PRESERVATION IN RECTAL CANCER: RESULTS ON A COHORT OF 70 PATIENTS

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Purpose/Objective: From Nov 1991 to March 2011, 70 patients (Pts) were entered in a cohort combining local excision (LE), brachytherapy (Ir192), and external beam radiotherapy (EBRT) for Pts at higher risk of nodal disease. Long term results are presented.

Materials and Methods: Seven Pts could not be treated as planned: 4 did not have a LE, and 3 did not have Ir192. There were 4 local recurrences (LR) in this subgroup and they are excluded from this analysis. There were 63 Pts treated as intended, 34 males and 29 females. Their age ranged from 33 to 88 with a median of 70, and 25 Pts (40%) were over 75 years old. There were 61 adenocarcinomas (18 WD, 35 MD, 6 PD, 2 unknown), one melanoma, and one neuroendocrine carcinoma. Tumor was located in the rectum in 34 Pts and at the anorectal junction in 29 Pts. Ir 192 dose was 45-50 Gy (Paris System) using single or double plane implant in the tumor bed and 20 Gy after EBRT. There were 21 T1, 33 T2, and 9 T3 Pts. Two of the T3 Pts had M1 disease. Ir 192 was used alone in 21 T1, and 20 T2 Pts. EBRT and Ir192 was used in 13 T2 and all T3 Pts. Chemotherapy was given with EBRT in 2 T2 and 7 T3 Pts.

Results: Following LE, margins were negative (R0) in 16 T1, 18 T2, 1 T3 and positive (R1) in 5 T1, 15 T2, and 8 T3 Pts. There were 14 LR, including one in the perirectal nodes: 4 T1 (19%), 8 T2 (24%), and 2 T3 (22%) Pts. There was no increased risk for T1 and T2 of LR with

regards to margins. There were 8 LR in the 34 R0 Pts (23%) and 4 LR in the 20 R1 Pts (20%). There were 6 salvage surgeries in 3 T1 and 2 T2 Pts: three are NED at 3, 7, 10 years, one is alive with disease at 3 years, and 2 died of metastatic disease. Two patients required a colostomy because of radionecrosis. Two patients had significant incontinence. There were 16 deaths from disease, including the two patients expected with Stage IV, and 7 died from other causes. **Conclusions:** This approach is promising for frail and elderly Pts at high surgical risk for a major resection. Adjuvant Ir192 is a valid alternative to major resection in some Pts with positive margins after LE.

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TRANSITIONAL NEOADJUVANT HIGH DOSE RATE ENDORECTAL BRACHYTHERAPY FOR RECTAL ADENOCARCINOMA

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Purpose/Objective: Neoadjuvant chemoradiation therapy (NCRT) is the standard of care for TxN1-2, T3-4N0, rectal adenocarcinoma. Previous studies have shown improved local control and reduced toxicity with NCRT compared to adjuvant therapy. NCRT consists of approximately 5-6 weeks of external beam radiation with concurrent 5-Fluorouracil. Intensity Modulated Radiation Therapy (IMRT) has been studied in attempt to further limit dosing and toxicity to surrounding normal tissue, but length of treatment does not differ. The addition of more aggressive chemotherapy to NCRT has increased acute toxicity rates, yet failed to improve response rates. Therefore a novel approach using high dose rate endorectal brachytherapy (HDR-EBT) given in 4 fractions (6.5Gy/day) on a pilot study was investigated. Preliminary results of the first seven patients on protocol were reviewed.

Materials and Methods: Cases of the first 7 patients enrolled on HDR-EBT pilot study, who underwent surgical resection, were reviewed. All patients had tumors <12 cm from the anal verge and no clinical/radiographic suspicious lymphadenopathy outside the mesorectum, T2-T3, N0-N1. All patients had NCI CTCAE toxicity assessments, MRI, PET/CT and CE pre/post HDR-EBT. RECIST/PERCIST were reported. Surgical specimens were reviewed by a single pathologist.

Results: 6 of the 7 patients had a decrease in CEA, 4-6 weeks post radiation, (mean 23%, range -4%-40%). All patients had a decrease in tumor size per MRI from average of 4 cm to 2.7cm. 6 of 7 pts had a decrease in the SUV max for post treatment PET/CT by >28%. Out of the 7 patients, 3 had node negative disease at resection, 2 patients had 1 positive node. All patients were margin negative and 6 had sphincter preserving surgery, (other patient chose to have APR due to age). 3 of 7 patients (43%) had a pathologic complete response (pCR) of their primary tumors. Toxicity assessments showed only 1 patient with grade 3 proctitis post-therapy, all others were grade 2 or less consisting of proctitis and pain that resolved following surgery. **Conclusions:** HDR-EBT is a novel approach for rectal cancer that reduces neoadjuvant treatment length, allows pts to undergo resection sooner, and shows favorable tumor response radiographically, pathologically, and biochemically. Preliminary results appear encouraging to guide future studies and assist with determining the efficacy of this modality.

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BRACHYTHERAPY WITH IODINE 125 OR RUTHENIUM 106 FOR TREATMENT OF CHOROIDAL MELANOMAS MEASURING 5-7 MM IN THICKNESS

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